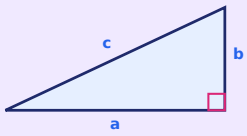
Unit 6 · Right Triangles and Trigonometry Cheat Sheet

Unit 6 focus: the Pythagorean Theorem, special right triangles and trig ratios

1 PYTHAGOREAN THEOREM

In a right triangle with legs a, b and hypotenuse c (opposite the right angle): $a^2 + b^2 = c^2$.

Example: Legs 7 and 24
Substitute: $49 + 576 = c^2$
Answer: $c = 25$



2 CONVERSE: CLASSIFY BY SIDES

Longest side c	Triangle
$c^2 = a^2 + b^2$	right
$c^2 < a^2 + b^2$	acute
$c^2 > a^2 + b^2$	obtuse

Sides 6, 8, 11: $121 > 36 + 64 = 100$, so obtuse. Common triples: 3-4-5, 5-12-13, 8-15-17.

3 SPECIAL RIGHT TRIANGLES

Triangle	Side ratio	Example
45°-45°-90°	$x : x : x\sqrt{2}$	leg 5 → hyp $5\sqrt{2}$
30°-60°-90°	$x : x\sqrt{3} : 2x$	short leg 4 → $4\sqrt{3}$, 8

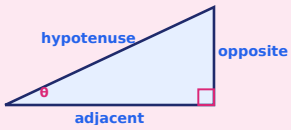
In a 30-60-90, the short leg is opposite 30° and the hypotenuse is twice it.

4 TRIG RATIOS: SOH-CAH-TOA

Ratio	Definition
$\sin\theta$	$\frac{\text{opposite}}{\text{hypotenuse}}$
$\cos\theta$	$\frac{\text{adjacent}}{\text{hypotenuse}}$
$\tan\theta$	$\frac{\text{opposite}}{\text{adjacent}}$

5 LABELING THE SIDES

- **Hypotenuse**: across from the right angle; always the longest.
- **Opposite**: across from the angle θ you are using.
- **Adjacent**: next to θ, not the hypotenuse.



6 QUICK REMINDERS & COMMON MISTAKES

✓ Do remember

- Calculator in **degree** mode.
- Label opposite, adjacent and hypotenuse from the angle you are using.
- Round only at the end.

✘ Common mistakes

- $a^2 + c^2 = b^2$: c must be the hypotenuse
- Opposite and adjacent swapped: they depend on the angle
- $\sin^{-1}$ to find a side: inverse trig finds angles

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More practice: finding sides and angles with SOH-CAH-TOA, elevation and depression

1 FINDING A SIDE WITH SINE

Opposite and hypotenuse → sine.

Example: $35°$, hypotenuse 12, find the opposite side x.
Ratio: $\sin 35° = \frac{x}{12}$
Solve: $x = 12 \sin 35°$
Answer: $x \approx 6.9$

2 FINDING A SIDE WITH COSINE

Adjacent and hypotenuse → cosine.

Example: $40°$, hypotenuse 15, find the adjacent side x.
Ratio: $\cos 40° = \frac{x}{15}$
Solve: $x = 15 \cos 40°$
Answer: $x \approx 11.5$

3 SIDE IN THE DENOMINATOR

When the unknown is on the bottom, multiply, then divide.

Example: $\tan 28° = \frac{9}{x}$
Rearrange: $x = \dfrac{9}{\tan 28°}$
Answer: $x \approx 16.9$

4 FINDING AN ANGLE (INVERSE TRIG)

Know two sides, want the angle: use $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$.

Example: Opposite 7, adjacent 24
Ratio: $\tan \theta = \frac{7}{24}$
Inverse: $\theta = \tan^{-1}\left(\frac{7}{24}\right)$
Answer: $\theta \approx 16.3°$

5 ELEVATION & DEPRESSION

Both angles are measured from a **horizontal** line. Draw the right triangle, then use SOH-CAH-TOA.

Example: From 50 ft away, the angle of elevation to a roof is $32°$. Height?
Ratio: $\tan 32° = \frac{h}{50}$
Solve: $h = 50 \tan 32°$
Answer: $h \approx 31.2$ **ft**

6 CHOOSING THE RATIO

Know or need opposite and hypotenuse	**sine**
Know or need adjacent and hypotenuse	**cosine**
Know or need opposite and adjacent	**tangent**
Know two sides, need the third (right triangle)	**Pythagorean Thm.**